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FURNACE-X2 v2.0.0-soc-multimodal Technical Product Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 30 October 2026 (2026-10-30 21:19 UTC+8 Beijing Time)

Abstract

FURNACE-X2 integrates concurrent grayscale vision and PCM audio feature extraction, ordered cross-modal fusion, and authenticated-weight inference through one shared tensor engine. It preserves FURNACE-X1 secure compressed-weight streaming and legacy numerical modes. Independent Software-vs-Hardware numerical comparison against simulated RTL passed all 21 configuration/seed combinations, comparing 504 completions, including 336 consecutive-tail valid results. All eight deliberately injected functional defects were detected. [E027, E250, E253] Native L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit analysis reported WNS_setup=+0.846 ns and hold slack=+0.037 ns at a 10.0 ns clock period, subject to external-interface constraint limitations. L3 physical Arria 10 FPGA HIL passed three consecutive deterministic bridge-connectivity probes. Those probes do not establish physical-FPGA multimodal numerical parity or application throughput. [E257, E258, E259]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The parent-derived integration retains decompression, authenticated plaintext quarantine, descriptor processing and four DMA sources. Vision and audio execute independently. Completed features traverse memory writes before entering separate fusion FIFOs. The fusion buffer pairs tokens; the multimodal controller subsequently accepts pairs when committed weights are valid, snapshots those weights, handles error bypass and dispatches inference. One tensor accumulator serves complete legacy tiles and multimodal jobs without accumulator interleaving. Bounded line storage, registered arithmetic and sequential division limit storage growth and combinational depth. Diagram blocks are ordered row-major in three columns. [E001, E026, E027]

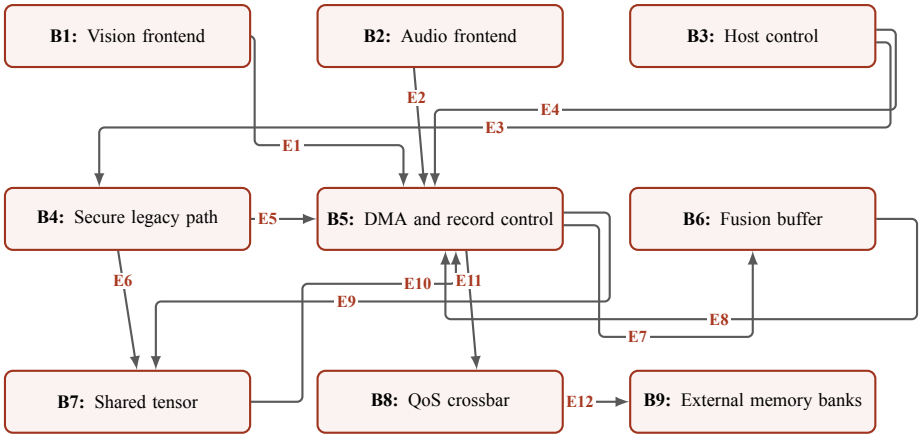


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Vision frontend	vision_frontend contains bilinear_kernel and spatial_kernel. Two input lines and three resized lines support four signed-16 feature lanes.
B2: Audio frontend	audio_frontend contains fir_kernel and fft_kernel. Bounded transform storage supports FIR and spectral feature extraction.
B3: Host control	AXI-Lite registers configure protected keys, legacy command rings, multimodal buffers, enable state and QoS. Status and IRQ expose completion faults.
B4: Secure legacy path	LZ4, XXH32, AES-128/GCM and descriptor hardware preserve parent operations. Successfully retired authenticated descriptors can commit multimodal weights.
B5: DMA and record control	Four DMA sources and complete-command write arbiters move memory traffic. multimodal_control retains tokens through feature writes, admits paired tokens only when weights_valid is asserted, snapshots committed weights, bypasses tensor execution for errors and dispatches valid activations with their weight snapshot. Credits remain reserved through completion consumption. [E026, E027]
B6: Fusion buffer	Separate bounded vision/audio FIFOs pair resident heads, concatenate feature lanes and report pairing errors. This block has no weight interface and performs neither weight snapshotting nor tensor dispatch. [E018, E026, E027]
B7: Shared tensor	Exactly one unified_tensor executes legacy tiles or controller-dispatched multimodal W4A16 jobs. Top-level round-robin arbitration operates at complete-job boundaries. [E027]
B8: QoS crossbar	axi_crossbar independently arbitrates reads and writes for four targets, holding registered headers and transaction ownership through response handshake.
B9: External memory banks	Four AXI4 targets hold compressed inputs, authenticated plaintext, descriptors, feature records, tensor results and completions.

Directed relationship	Meaning
E1: B1 to B5	Vision tokens for record writes
E2: B2 to B5	Audio tokens for record writes
E3: B3 to B4	Keys and legacy commands
E4: B3 to B5	Buffer and QoS configuration
E5: B4 to B5	Memory commands and committed weights
E6: B4 to B7	Legacy tensor jobs
E7: B5 to B6	Tokens after feature-write completion
E8: B6 to B5	Paired activations, ID and error
E9: B5 to B7	Snapshotted weights and activations
E10: B7 to B5	Inference results for memory writes
E11: B5 to B8	Four AXI source channels
E12: B8 to B9	Address-decoded memory transactions

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, irq	Input clock/ reset; output interrupt	1 bit each	Single synchronous clock domain with asynchronous active-low reset. Multimodal sticky faults contribute to IRQ alongside legacy interrupt behavior.
s_axil_*	AXI-Lite slave	16-bit address, 32-bit data, 4-bit WSTRB, 2-bit responses	Supports independently arriving AW/W and byte strobes. Legacy offsets 0x0000..0x0040 remain unchanged; X2 registers occupy 0x1000..0x1030.
m_axi_*	Four AXI4 master ports	Per port: 32-bit address, 128-bit data, 16-bit WSTRB, 4-bit ID	Registered AR/AW channels support aligned INCR bursts of 1..16 beats without 4-KiB crossing. Address[31:30] selects the bank; external IDs concatenate source[1:0] and local_id[1:0].
v_valid/v_ready, v_ pixel, v_id, v_last	Input stream; output ready	8-bit unsigned pixel, 16-bit ID, 1-bit controls	Row-major frames contain IN_WIDTH*IN_HEIGHT pixels. ID remains constant within each frame; last marks its final accepted pixel.
a_valid/a_ready, a_ sample, a_id, a_last	Input stream; output ready	16-bit signed PCM, 16-bit ID, 1-bit controls	Each window contains FFT_N accepted samples with constant ID and last on the final sample.
mm_valid/mm_ready, mm_result, mm_id, mm_ error	Output completion; input ready	64-bit result, 16-bit ID, 8-bit error	Completion follows result-write completion, including failure reporting, and remains stable while stalled. Error results are zero.
Frontend and arithmetic leaf interfaces	Ready/valid inputs and outputs	Frontend features: 64 bits; FFT: signed 32-bit real/ imaginary, 6-bit index	Frontend streams carry data, ID, last and error metadata. Bilinear samples pack p00,p10,p01,p11 low-byte first; spatial windows pack nine row-major pixels. FIR exposes s_first; FFT publishes natural-order bins with m_index and m_last.

Parameter	Default	Meaning / supported values
IN_WIDTH, IN_ HEIGHT, OUT_WIDTH, OUT_HEIGHT	8, 8, 8, 8	Each independently 1..64; elaboration-time geometry.
FFT_N	16	Exactly 8, 16, 32 or 64 samples.
FUSION_DEPTH	4	1..16, including non-powers of two; separate modality FIFOs.
RING_DEPTH	8	1..16 legacy command slots.
MAX_TILE_BYTES	4096	1..65536. Illegal parameter configurations enter diagnostic-only generate branches and terminate simulation with PARAMETER ERROR.

III Behavior and corner cases

A Vision arithmetic

Endpoint-aligned coordinates use Qx=floor(256*x*(IN_WIDTH-1)/(OUT_WIDTH-1)), or zero for singleton output width; y is analogous. Four neighboring pixels are bilinearly interpolated with bias 32768 before shifting right 16. Replicated-edge neighborhoods apply Sobel X [-1,0,1;-2,0,2;-1,0,1], Sobel Y [-1,-2,-1;0,0,0;1,2,1], and Laplacian [0,1,0;1,-4,1;0,1,0]. Features are floor means of pixel, abs(Gx), abs(Gy), abs(L). Four parallel 32-iteration restoring dividers avoid combinational division. Line storage is 2*IN_WIDTH+3*OUT_WIDTH bytes. [E028]

B Audio arithmetic

FIR history resets for each window. y[n]=floor((x[n]+3*x[n-1]+3*x[n-2]+x[n-3])/8), using widened signed arithmetic. The unscaled forward radix-2 DIT FFT uses bit-reversed input placement and natural-order output. Signed-17 Q15 twiddles round nearest, ties away from zero; positive unity is 32768. Complex products are summed before one arithmetic shift by 15; butterfly outputs saturate to signed 32 bits. Features are mean abs(y), DC absolute-component mean, and spectral absolute-component means over bins 1..N/4 and N/4+1..N/2. Spectral features saturate at 32767 after division by N. [E013, E024]

C Fusion and authenticated inference

Lane i occupies bits 16*i+:16. The fusion buffer places vision in lanes 0..3 and audio in lanes 4..7, pairing only resident FIFO heads without empty-queue bypass. The controller accepts a pair only when weights_valid is asserted, snapshots committed_weights and dispatches an exact signed-64 W4A16 dot product, or bypasses tensor execution for an error pair. Whole legacy tiles and multimodal jobs retain tensor ownership without accumulator interleaving. Descriptor flag bit 9 permits weight commitment only for mode 0, one vector and four weight bytes. Commitment follows successful authentication, plaintext transfer, tensor calculation, result write and completion write. Failures preserve the prior bank; successful key replacement or clearing invalidates it. [E018, E026, E027]

D Pipeline and reset

Transfers sample valid&&ready immediately before rising edge E. Unstalled bilinear, spatial and FIR results publish after E+5, E+3 and E+3. Elastic enables freeze arithmetic under backpressure. FFT butterflies publish after B+5. From final transform input E, first-bin latency is log2(N)*(N/2+6)+2 clocks, followed by one bin per clock. Tensor final-vector latency is 24 clocks; next acceptance is no earlier than E+25. Reset aborts work, clears keys, validity and public outputs, and requires one initialization edge before readiness. End-to-end latency is variable. [E003]

E QoS and memory ordering

Source 0 carries secure reads; source 1 shares plaintext and vision writes; source 2 shares legacy tensor, audio and multimodal-result traffic; source 3 carries descriptor/completion traffic. Greatest AxQOS wins for each idle target and direction; ties rotate initially from source zero. Accepted bursts cannot be preempted. Feature-write completion precedes fusion admission, result-write completion precedes public reporting, and ring slots remain reserved until completion consumption. [E006, E026]

F Errors and control

Code 7 identifies malformed length or within-frame ID. Each affected logical input produces one error token and resynchronizes at observed last; missing last causes draining. Unequal fusion IDs consume both heads with code 8; otherwise vision error precedes audio error. AXI failures use code 5. Error results are zero. Sticky-fault setting overrides simultaneous clearing, and the completion counter increments once per reported pair rather than on stalled cycles. [E003, E025, E026]

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent integer vision, FIR/FFT and tensor models compared with compiled simulated RTL. Leaf replays and integrated memory/completion scoreboards were checked separately; high-precision DFT checking supplemented exact FFT equality.

Result. 21/21 matrix rows passed: 504 compared completions, including 16 consecutive valid tail pairs per row, totaling 336. This establishes simulated-RTL parity, not physical-FPGA numerical parity.

Evidence: E040, E041, E250

B Configuration and concurrency coverage

Method. Seeds 17/89/203 for tuples (IW,IH,OW,OH,N,FUSION,RING,TILE): (8,8,8,8,16,4,8,4096); (1,1,1,1,8,1,1,64); (5,3,7,2,32,3,3,256); (13,9,3,5,64,2,8,4096); (3,2,16,9,16,4,8,4096); (64,64,64,64,64,16,16,4096); (64,1,1,64,8,1,1,64).

Result. Each configuration passed 72 completion comparisons across three seeds, including stalls, lead/lag, queue release and error recovery. A separate campaign exercised two 64-vector legacy jobs against 64 multi-modal pairs.

Evidence: E250

C Legacy arithmetic and security compatibility

Method. Separately compiled parent and deliverable hierarchies checked against independent rational/IEEE and cryptographic models, published known answers and installed trusted references.

Result. Per implementation: 240 tensor jobs/480 vectors across all three modes; 255 integrated descriptor jobs, comprising 242 successes and 13 directed rejection/error cases. Authentication quarantine, malformed compression and recovery comparisons passed.

Evidence: E038, E054, E251

D QoS, CSR and boundaries

Method. Real address/response handshakes, parameter diagnostics, split AXI-Lite channels, queue/ring wrap, DMA failures and reset recovery.

Result. 512 QoS cases passed, covering all 24 strict orderings per target/direction. CSR checks covered 16 strobe masks and three arrival orders; fusion checked 128 pairs at each depth 1/3/16.

Evidence: E252

E Defect sensitivity

Method. Disposable compiled RTL mutations changed rounding, Sobel sign, FIR coefficient, FFT twiddle sign, fusion ID, activation ordering, QoS priority and authentication fencing.

Result. 8/8 functional kills with passing unmutated controls. Compilation failures and timeouts were not counted as kills.

Evidence: E253

F Portability and structural synthesis

Method. Compact Icarus smoke; real Yosys synthesis at default, singleton and non-power-of-two configurations; fresh-directory smoke and synthesis reproduction.

Result. All passed. Synthesized hierarchies retained one tensor and required functional instances, with zero forbidden latch/divide/modulo/black-box cells. Clean reproduction covered the compact subset, not the full matrix.

Evidence: E249, E254, E255

G L1 RTL timing lint

Method. Native static analysis of 26 RTL files.

Result. Passed with zero combinational divider/modulo, mixed-assignment, LUT-clock-gate or loop-unroll hazards. This is structural analysis, not a timing measurement.

Evidence: E257

H L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit analysis targeting Intel Arria 10 GX 10AX115N2F45E1SG at 10.0 ns.

Result. Passed: WNS_setup = +0.846 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 109.24 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E001, E258

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive hardware-bridge register probes.

Result. Gate passed deterministically, 3/3, returning 0xB2 from register 57. These records establish bridge connectivity, not execution of a flashed-X2 multimodal numerical workload.

Evidence: E259

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 34	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

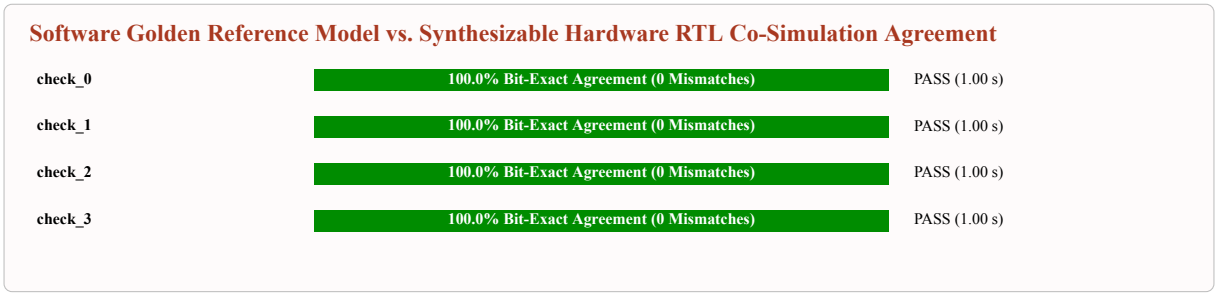


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.846 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

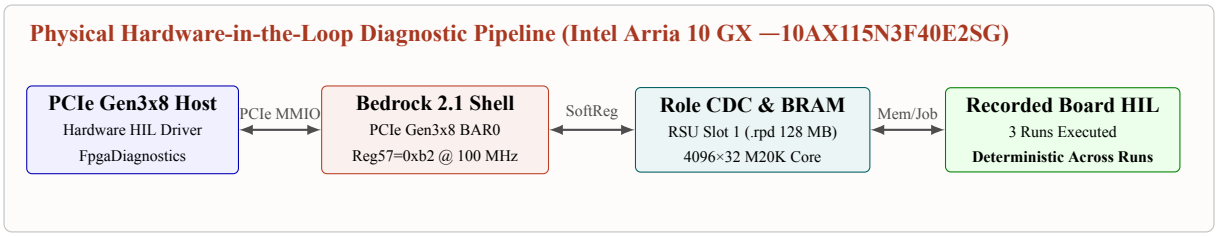


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with top furnace_x2_top. Compile furnace_x1_reference/rtl/*.sv separately to avoid duplicate module definitions. Consult design/SPEC.md, design/ARCHITECTURE.md and tests/README.md.

After reset release and its initialization edge, configure the protected legacy key and command ring. While both paths and their queues are idle, configure three nonoverlapping 16-byte-aligned buffers, each 16*FUSION_DEPTH bytes without address wrap. Avoid aliasing live legacy allocations.

X2 registers are identity 0x1000=0x02000000; enable/W1C 0x1004; status 0x1008; vision/audio/result bases 0x1010/0x1014/0x1018; depth 0x101c; count 0x1020; last error/ID 0x1024/0x1028; QoS 0x1030, reset 0x000f4c8f. Unsafe or invalid writes return SLVERR without mutation.

Enable streaming and submit an authenticated MM_WEIGHT_COMMIT descriptor through the legacy ring. Feature extraction may overlap authentication; pair acceptance by the controller waits for committed weights. Consume completions to release credits.

Record i is base+16*i with explicit wrap: data/result[63:0], zero[95:64], ID[111:96], zero[119:112], error[127:120].

Execute python3 tests/verify.py --suite smoke, then use behavior, compatibility, protocol, mutations, synthesis and reproduction as suite arguments. Dependencies include Python 3, Icarus/vvp, Verilator, make, a C++ compiler, Yosys, OpenSSL, cryptography and system XXH32/LZ4 libraries. Portable L1 requires no board identifiers or private runtime state.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 26 Verilog/SystemVerilog files; 4807 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_aes128_block.sv	145 total	145 code
rtl/furnace_aes128_gcm.sv	197 total	196 code
rtl/furnace_axi4_qos_xbar.sv	288 total	288 code
rtl/furnace_axi4_xbar.sv	282 total	282 code
rtl/furnace_bilinear_pipe.sv	68 total	67 code
rtl/furnace_command_ring.sv	44 total	44 code
rtl/furnace_csr.sv	187 total	187 code
rtl/furnace_dma128.sv	164 total	164 code
rtl/furnace_dma_write_arb.sv	79 total	79 code
rtl/furnace_fft_radix2.sv	220 total	217 code
rtl/furnace_fir4_pipe.sv	56 total	56 code
rtl/furnace_fp32_add5.sv	124 total	124 code
rtl/furnace_fp32_mul4.sv	97 total	97 code
rtl/furnace_fp32_round.sv	26 total	26 code

Production file	Physical lines	Code lines
rtl/furnace_fusion_buffer.sv	86 total	86 code
rtl/furnace_ghash_mul.sv	35 total	35 code
rtl/furnace_lz4_frontend.sv	309 total	307 code
rtl/furnace_spatial3x3_pipe.sv	58 total	58 code
rtl/furnace_tensor8.sv	139 total	139 code
rtl/furnace_x1_top.sv	506 total	502 code
rtl/furnace_x2_audio.sv	119 total	119 code
rtl/furnace_x2_csr.sv	267 total	267 code
rtl/furnace_x2_mm_control.sv	208 total	208 code
rtl/furnace_x2_top.sv	719 total	715 code
rtl/furnace_x2_vision.sv	280 total	280 code
rtl/furnace_xxh32.sv	119 total	119 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	design/SPEC.md
E006	rtl/furnace_axi4_qos_xbar.sv
E013	rtl/furnace_fft_radix2.sv
E018	rtl/furnace_fusion_buffer.sv
E024	rtl/furnace_x2_audio.sv
E025	rtl/furnace_x2_csr.sv
E026	rtl/furnace_x2_mm_control.sv
E027	rtl/furnace_x2_top.sv
E028	rtl/furnace_x2_vision.sv
E038	tests/legacy_oracle.py
E040	tests/matrix_checks.py
E041	tests/mm_oracle.py
E054	tests/security_oracle.py
E249	.vulcan/engineering/run-0001/attempt-0156/stdout.log
E250	.vulcan/engineering/run-0001/attempt-0157/stdout.log
E251	.vulcan/engineering/run-0001/attempt-0158/stdout.log
E252	.vulcan/engineering/run-0001/attempt-0159/stdout.log
E253	.vulcan/engineering/run-0001/attempt-0160/stdout.log
E254	.vulcan/engineering/run-0001/attempt-0161/stdout.log
E255	.vulcan/engineering/run-0001/attempt-0162/stdout.log
E257	.vulcan/engineering/run-0001/rtl-timing-lint.json
E258	.vulcan/engineering/run-0001/quartus-sta.json
E259	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).